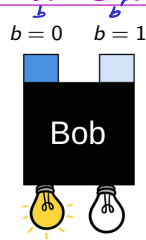
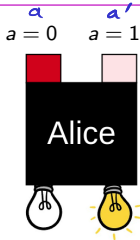


Bell inequality: "Device-independent" quantum cryptography

$$N_{\text{total}} = N_{++} + N_{--} + N_{+-} + N_{-+}$$



$A = +1$ $A = -1$

$B = +1$ $B = -1$

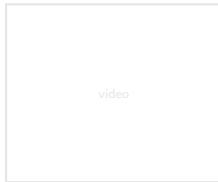
Data:

a-meas	+1	+1	-1	-1	-1	...
b-meas	-1	+1	-1	+1	+1	...

Correlation: $\mathcal{C}(a, b) = \frac{N(++|ab) + N(--|ab) - N(+-|ab) - N(-+|ab)}{N_{\text{total}}}$

CHSH Inequality: $\mathcal{C}(a, b) + \mathcal{C}(a', b) + \mathcal{C}(a, b') - \mathcal{C}(a', b') \leq 2$

Violation of a Bell inequality guarantees that the observed outputs are **not prerecorded** and they arise from entangled quantum systems that possess **intrinsic randomness**, which can be use for **secure communication**.



$Q\pi_{AI}$ Bell inequality: the question is about the origin of probabilistic nature

Bell's inequality does not apply to the quantum correlations, which means that one or more of its assumptions do not hold true for quantum physics.

- **Locality:** The **result** of a measurement on a subsystem must be unaffected by **operations** on the other subsystem that is **far far away** (spatially separated).

$C(a,b)$ "In a theory in which parameters are added to quantum mechanics to determine the results of individual measurements, without changing the statistical predictions, there must be a mechanism whereby the setting of one measuring device can influence the reading of another instrument, however remote."

"Local" HVT
do not
satisfy
QM predictions

- **Realism (hidden determinism):** Physical properties (such as spin components, position, momentum, energy) have definite values, which exist independent of observation.

- Most physicists believe that the measurement outcomes are **fundamentally (intrinsically) random**, and there is **no hidden determinism**.
- Violation of a Bell inequality guarantees that the observed outputs are **not prerecorded** and they arise from entangled quantum systems that possess **intrinsic randomness**, which can be used for **secure communication** (Device-independent quantum cryptography).

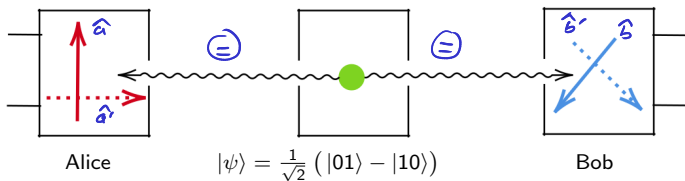
J. S. Bell, Physics 1, 195 (1964).

R. Colbeck, arXiv:0911.3814 [quant-ph], Chapt 5.

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reem, Nature **464**, 1021 (2010).

Q π_{AI} Bell inequality: the question is about the origin of probabilistic nature



$$c(\hat{a}, \hat{b})$$

Quantum correlation: $\langle \psi | \hat{a} \cdot \vec{\sigma} \otimes \hat{b} \cdot \vec{\sigma} | \psi \rangle = -\hat{a} \cdot \hat{b} = -\cos \theta$ (Born rule)

CHSH Inequality: $\underbrace{c(a, b)} + \underbrace{c(a', b)} + \underbrace{c(a, b')} - \underbrace{c(a', b')} \leq 2$

$$2.8 \approx 2\sqrt{2} = 4\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$

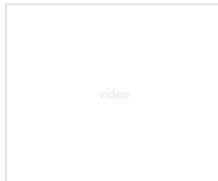
Bell's inequality does not apply to the quantum correlations, and its violation is an established experimental fact.

$$\hat{a} = \hat{z}$$

$$\hat{a}' = \hat{x}$$

$$\hat{b} = \frac{-\hat{x} - \hat{z}}{\sqrt{2}}$$

$$\hat{b}' = \frac{\hat{x} - \hat{z}}{\sqrt{2}}$$

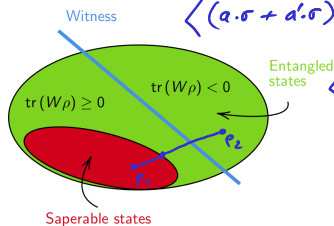


LOCC, Entanglement witness, and Entanglement measure

$$w\rho_1 + (1-w)\rho_2 = \rho \quad 0 \leq w \leq 1$$

Convex

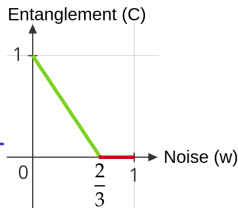
- Entanglement does not increase under Local Operations and Classical Communication, $\rho \rightarrow U_a \otimes U_b(\rho)U_a^\dagger \otimes U_b^\dagger$.
- A Bell inequality is an entanglement witnesses.



$$\langle (a \cdot \sigma + a' \cdot \sigma) \otimes b \cdot \sigma + (a \cdot \sigma - a' \cdot \sigma) \otimes b' \cdot \sigma \rangle \leq 2$$

$$0 \leq 2 - \langle \dots \rangle$$

$$\langle (A+A')B + (A-A')B' \rangle \leq 2$$



$$\frac{I}{4} = \frac{1}{4} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Werner state $\rho = w \underbrace{\frac{I}{4}}_{\rho_1} + (1-w) \underbrace{|\psi\rangle\langle\psi|}_{\rho_2}$

Completely mixed state $\frac{I}{4}$

Maximally entangled state $|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$

- The concurrence $C := \max\{0, \sqrt{\lambda_m} - \sum_{\lambda \neq \lambda_m} \sqrt{\lambda}\}$ measures two-body entanglement where $\lambda \geq 0$ are the eigenvalues of $\rho \tilde{\rho}$, $\tilde{\rho} = (Y \otimes Y)\rho^*(Y \otimes Y)$, ρ^* is the complex conjugate of ρ , and $\lambda_m = \max\{\lambda\}$.

S. Hill and W.K. Wootters, Phys. Rev. Lett. **78**, 5022 (1997). W. K. Wootters, Phys. Rev. Lett. **80**, 2245 (1998).
R. F. Werner, Phys. Rev. A **40**, 4277 (1989). R. Horodecki, et al., Rev. Mod. Phys. **81**, 865 (2009).